Coordinates & Transformations

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: A point A is at (3, 4) on a coordinate grid. What is its x-coordinate and what is its y-coordinate?

Answer: The x-coordinate is given first: 3. The y-coordinate is given second: 4.

Standard: A point B is at (2, 5). It is translated 3 right and 1 up. What are its new coordinates?

Answer: 3 right: 2 + 3 = 5. 1 up: 5 + 1 = 6. New point: (5, 6).

Advanced: A point C is at (4, 3). It is reflected in the x-axis. What are its new coordinates?

Answer: Reflecting in the x-axis keeps x the same and reverses the sign of y: (4, −3).

Easy / Standard (15 questions)

1. A point A is at (2, 5). What is its x- coordinate?
2. A point A is at (2, 5). What is its y- coordinate?
3. A point starts at (1, 2) and is translated 3 right. What are its new coordinates?
4. A point starts at (2, 1) and is translated 4 up. What are its new coordinates?
5. A point is at (3, 2). It is translated 2 right. What are its new coordinates? A) (5, 2) B) (3, 4) C) (1, 2) D) (5, 4)
6. Point B is at (0, 3). Which axis does it lie on?
7. Point C is at (6, 0). Which axis does it lie on?
8. A point starts at (4, 3) and is translated 1 right and 2 up. What are its new coordinates?
9. A point starts at (5, 5) and is translated 4 left. What are its new coordinates?
10. Select ALL that apply. Look at the points P(3, 3), Q(3, 6), R(7, 3) and S(7, 6). Which have an x- coordinate of 3? A) P(3, 3) B) Q(3, 6) C) R(7, 3) D) S(7, 6)
11. A point starts at (0, 0) and is translated 6 right and 4 up. What are its new coordinates?
12. A point starts at (4, 4) and is translated 3 down. What are its new coordinates? A) (4, 1) B) (1, 4) C) (7, 4) D) (4, 7)
13. A point D is at (2, 2) and a point E is at (2, 7). Are D and E on the same vertical line or the same horizontal line?
14. A point starts at (5, 1) and is translated 2 right and 3 up. What are its new coordinates?
15. Which of these points lies exactly on the x-axis? A) (4, 0) B) (0, 6) C) (2, 2) D) (5, 5)

Standard (15 questions)

1. A point starts at (3, 4) and is translated 5 right and 2 up. What are its new coordinates?
2. A point is at (4, 3). It is reflected in the x-axis. What are its new coordinates?
3. A point starts at (5, 6) and is translated 4 down. What are its new coordinates?

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4. A point at (2, 6) is reflected in the x-axis. What are its new coordinates? A) (2, -6) B) (-2, 6) C) (6, 2) D) (-2, -6)

5. A point is at (3, 5). It is reflected in the y-axis. What are its new coordinates?

6. A rectangle has vertices at (1, 1), (5, 1), (5, 4) and (1, 4). What is the length of its longer side?

7. A point starts at (6, 2) and is translated 1 right and 5 up. What are its new coordinates?

8. A point starts at (4, 8) and is translated 6 left. What are its new coordinates?

9. A point at (5, 2) is reflected in the y-axis. What are its new coordinates? A) (-5, 2) B) (5, -2) C) (2, 5) D) (-5, -2)

10. A point starts at (0, 7) and is translated 3 right and 2 down. What are its new coordinates?

11. Select ALL that apply. A square has vertices at (2, 2), (2, 6), (6, 6) and (6, 2). Which statements are true? A) The square has a side length of 4 units B) The square has a side length of 6 units C) The point (6, 6) is a vertex of the square D) The point (2, 6) is a vertex of the square

12. A point starts at (2, 3) and is translated 4 right and 4 up. What are its new coordinates?

13. A point is at (7, 1). It is reflected in the x-axis. What are its new coordinates?

14. A point starts at (3, 3) and is translated 2 left and 5 up. What are its new coordinates? A) (1, 8) B) (5, 8) C) (1, -2) D) (5, -2)

15. A point starts at (8, 0) and is translated 2 up. What are its new coordinates?

Standard / Hard (15 questions)

1. Starting at (7, 2), a point is reflected in the y-axis. Where does it end up?

2. Starting at (6, 9), a point is reflected in the x-axis this time. Where does it end up?

3. A point at (5, 3) is reflected in the x-axis. What are its new coordinates? A) (5, -3) B) (-5, 3) C) (-5, -3) D) (3, 5)

4. A rectangle has vertices at (0, 0), (7, 0), (7, 4) and (0, 4). What are the coordinates of the vertex diagonally opposite (0, 0)?

5. A point at (8, 6) is reflected in the y-axis. What are its new coordinates? A) (-8, 6) B) (8, -6) C) (-8, -6) D) (6, 8)

6. From a starting position of (2, 9), a point is moved 6 units right and 7 units down. Give its new coordinates.

7. A square has one vertex at (0, 0) and the diagonally opposite vertex at (6, 6), with sides parallel to the axes. What are the coordinates of the other two vertices?

8. Select ALL that apply. A shape has vertices A(2, 1), B(2, 6), C(9, 6) and D(9, 1). Which sides have a length of 5 units? A) Side AB B) Side BC C) Side CD D) Side DA

9. Beginning at (1, 1), a point is shifted 8 units to the right and 6 units upward. What position does it reach?

10. A point is at (9, 4). It is reflected in the y-axis. What are its new coordinates?

11. A rectangle has vertices at (0, 0), (9, 0), (9, 5) and (0, 5). What is the perimeter of the rectangle?

12. A point starts at (3, 8) and is translated 5 right and 6 down. What are its new coordinates? A) (8, 2) B) (8, 14) C) (-2, 2) D) (-2, 14)

13. A point is at (2, 2). It is translated so that it lands exactly on the y-axis, moving only left. How far left did it move?

14. A point at (4, 7) is reflected in the x-axis to give point P. What is the y-coordinate of P?

15. A point is at (3, 2). It is translated 4 up, then reflected in the x-axis. What are its final coordinates?

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